

# Klootwijk SDF

## Directional Definition Calculus

### Consolidation D1

Individual definitions.

Packed composition.

Causal throughput.

The mathematical formalization and its  
present-day and far-future possibility space.

A definition supplies a direction, not an intrinsic length or a built-in inside or outside. Its meaning, its relationships and its change over time are defined explicitly. Packing keeps the individual definitions recoverable.

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A new mathematical consolidation of the stated project direction.

Plain-language explanations accompany the definitions and examples.

## The consolidation in one page

**Your goal is a common way to define things, operations, expressions, relationships and change, with each definition retaining its own identity.** Your latest clarification places a directional eigenaxis at the foundation. It does not place a sampled spatial volume, a signed distance, a magnitude, or an inside/outside test there.[\[P0, H\]](#)

This edition calls that direction the **Klootwijk SDF Directional Definition Calculus**. The project label *SDF* is retained. The descriptive name is introduced here; it is not presented as a recovered title from an unavailable original.

### In everyday language

Imagine a collection of named arrows. Stretching an arrow does not change which way it points. Each arrow has its own definition. Some definitions name things; some say how things relate; some say what can happen next. Putting them in one collection does not melt them into one giant object. You can still point to each definition, change it, reuse it and follow what depends on it.

**What is formalized.** The document defines direction-only primitives, their eigenaxis witnesses, individual identity, lossless packing, operators represented in the same notation, composition, explicit time, self-reference, optional measurements, geometric and topological descriptions, and throughput from inputs to justified outcomes.

**What the possibility space becomes.** A single descriptive calculus can organize route problems, design constraints, changing systems, mechanism models and learned rules. More ambitious extensions include self-repairing digital structures, adaptive materials, molecular rule systems, robotic collectives, virtual tissues and entirely invented mathematical worlds. The connection to each application is a stated construction, not a claim that naming an application already realizes it.

**What changed from the previous reports.** Their signed-distance and Boolean-room constructions are not the foundation here. Their implementation-centered framing is also not the foundation. What survives is the unification objective: individually defined things and rules, reusable composition, explicit change and broad applications.[\[P1, P2\]](#)

**Source basis.** The two supplied consolidation PDFs have been inspected, alongside the visible project history and your current clarification. The original geometry, routing and nature source PDFs themselves remain unavailable. New mathematics supplied to reconcile the requirements is identified as D1 construction; it is not silently attributed to those missing documents. Appendix B gives the precise source ledger.

**The central result:** a dimension-independent, direction-only definition calculus with explicit composition and causal semantics. Its strength is the ability to state and connect many kinds of mechanism without making intrinsic magnitude or containment part of every primitive.

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**Reading routes.** For the idea without the mathematics, read Sections 1–2, the blue explanation boxes, and Sections 12–16. For the exact formalization, read Sections 3–11 and the compact charter in Appendix A. Appendix B records the source lineage and what was actually available.

# 1 Your goal and the revised foundation

## 1.1 What the revisions were reaching toward

The visible sequence begins with geometric definitions and named shapes. It expands into operators and expressions being definitions themselves, compact individual representations, logarithmic/polar organization, an explicit time operator, self-reference, and applications involving routes and mechanisms found in nature. Your later requests repeatedly reject a mere calculator or renderer as the destination.[\[H, P1, P2\]](#)

The common ambition is **representational unity without losing individuality**: the thing being described, the rule relating it to something else, and the rule governing its change should all be expressible within the same formal vocabulary. A relationship should not need to become an unrelated kind of object merely because it is an operation rather than a shape.

Your current clarification adds a decisive constraint: the native SDF supplies a directional eigenaxis, not an inherent scalar distance, length or containment classification. D1 treats that as the current definition of the project primitive.[\[P0\]](#)

## 1.2 A precise terminology boundary

In conventional mathematical and geometric usage, a signed-distance function returns a signed scalar distance to a boundary. An eigenvector instead concerns a linear operator that maps a nonzero vector to a scalar multiple of itself. These are different mathematical objects.[\[R1, R2\]](#)

Here, **SDF means the project's direction-only definition primitive**. When conventional signed distance is needed for a particular application, it must be introduced as an additional, explicitly named measurement. Nothing in the native calculus assumes a cubic grid, three or four spatial coordinates, or a pre-existing volume. The distinction is the stated mathematical contract, not an assertion that conventional mathematics is limited to cubic grids.

## 1.3 What D1 retains and what it replaces

| Theme                   | Retained                                             | D1 refinement                                                                     |
|-------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------|
| Individual definitions  | Objects, rules and expressions have definitions.     | Each has a recoverable identity and a directional representation.                 |
| Packing and reuse       | Many definitions belong together.                    | Packing preserves components; it is not vector addition or spatial fusion.        |
| Time and self-reference | Change and references are first-class.               | Time is an explicit transition relation; cyclic references have stated semantics. |
| Geometry and nature     | Shape, routes and mechanisms remain in scope.        | Physical properties are separate definitions, not inferred from a bare axis.      |
| Throughput              | Definitions are intended to produce useful outcomes. | Throughput is defined relationally and causally, independently of a device.       |

**In everyday language**

The earlier reports began with little rooms and asked whether an answer was inside a room. This version begins with individually named directions and asks what each direction means and which relationships connect them. A room can still be described, but rooms are no longer the starting assumption.

## 2 The objects of the calculus, before equations

A complete definition has an **identity**, a **declared role** and a **directional expression**. Identity tells us which definition is being referenced. The role tells us whether its interpretation is a value, a relationship, a clock rule, a measurement, a shape description or a description of another definition.

The **native directional content** has no intrinsic length. Its interpretation can nevertheless refer to a length, just as the written word “metre” is not itself one metre long. This distinction is essential for keeping the primitive direction-only while still allowing quantitative applications.

A **pack** is an organized collection of such definitions together with explicit connections. The connections are also describable as definitions. A **composition** connects an output role to a compatible input role. A **transition** relates an earlier situation to a later one. An **observation** specifies what counts as an answer to a question.

**Direction is the native representation. Interpretation supplies meaning. Composition supplies relationships. Time supplies allowed change. Measurement supplies quantities only where a model explicitly calls for them.**

This is not merely a change of names for the earlier signed-distance construction. The mathematical equivalence relation, the meaning of packing, the status of time, and the conditions for physical interpretation are all stated afresh below.

## 3 The individual SDF: an oriented eigenaxis

### 3.1 Direction without intrinsic magnitude

Choose a real vector space  $V$  for a particular kind of definition. It is an abstract carrier, not necessarily physical space. Different kinds may use different carriers. No universal spatial dimension is imposed.

**Definition 3.1** (Oriented directional content). For nonzero vectors  $v, w \in V$ , define

$$v \sim_+ w \iff w = av \text{ for some } a > 0. \quad (1)$$

The oriented direction represented by  $v$  is  $[v]_+$ . The collection of such directions is

$$\text{Dir}(V) = (V \setminus \{0\}) / \sim_+.$$

There is no chosen vector length in this definition. For example,  $(1, 2)$ ,  $(3, 6)$  and  $(10, 20)$  represent the same directional content. Their coordinate ratios agree, but their representative lengths need not. A norm is not needed to define the quotient.

**Orientation matters.** The directions  $[v]_+$  and  $[-v]_+$  are distinct. An unoriented axis would instead identify both signs. D1 uses the oriented version because your wording includes direction and because a one-bit distinction can then be represented by opposite orientations. This is an explicit D1 choice, not a claim that the missing PDFs had already settled that convention.

**Zero is not a direction.** A zero vector cannot provide an axis. A definition that has no directional result at a particular argument is undefined there, or returns an explicitly defined status such as “stationary.” The status is a separate definition; it is not a hidden zero-length arrow.

#### In everyday language

A small arrow pointing northeast and a long arrow pointing northeast mean the same direction. Turning the arrow around changes its direction. Erasing the arrow entirely does not create a special direction; it leaves no arrow.

### 3.2 Making “eigenaxis” literal

An eigenaxis must be associated with an operator. An arbitrary arrow is not, by itself, evidence of a natural or physical eigenmode. The standard relation is  $Lv = \lambda v$  for  $v \neq 0$ .[\[R2\]](#)

**Definition 3.2** (Eigenaxis witness). A directional SDF may have a witness  $L : V \rightarrow V$  such that

$$Lv = \lambda v, \quad \lambda > 0. \quad (2)$$

The native result remains  $[v]_+$ ;  $L$  and  $\lambda$  belong to the explicit witness description, not to an intrinsic length of the result.

The positive eigenvalue preserves the oriented direction. A negative eigenvalue preserves the underlying line but reverses orientation. Those two situations must not be conflated. A repeated eigenvalue can also admit many directions; it does not automatically select one. For example, the identity operator preserves every direction. A real planar quarter-turn has no nonzero real eigenvector. The choice of carrier and witness therefore matters.

**Proposition 3.1** (Every direction admits an eigenaxis witness). For every nonzero  $v$  in a finite-dimensional real vector space, there is a linear map  $P$  with  $P^2 = P$  and  $Pv = v$ .

*Proof.* Choose a linear functional  $\ell$  satisfying  $\ell(v) = 1$ , and set  $P(w) = v\ell(w)$ . Then  $Pv = v$  and  $P^2(w) = v\ell(v\ell(w)) = v\ell(w) = P(w)$ . Replacing  $v$  by  $av$  and  $\ell$  by  $\ell/a$  leaves  $P$  unchanged.  $\square$

This construction makes the word eigenaxis exact without attaching a physical amplitude. The eigenvalue 1 is a witness convention, not a law of nature. Many different witnesses can preserve the same direction. Consequently, knowing a direction alone does not recover a unique operator, material law or mechanism.

### 3.3 Named definitions and contextual results

A readable definition record is written

$$D_i = (i, \tau_i, e_i), \quad (3)$$

where  $i$  is a unique identity,  $\tau_i$  declares the role or input/output types, and  $e_i$  is its finite expression. These are descriptive slots in a record, not spatial coordinates. Section 4 gives a single direction-only representation of the entire record.

For a single-result definition applied to a permitted context  $c$ , the native result, when defined, is

$$\text{SDF}_i(c) = [v_i(c)]_+ . \quad (4)$$

A context can be a symbolic expression, a collection of named arguments, an event history or a geometric query. It need not be a physical location. A relation that admits several results returns a collection of individually identified directional answers, not one fabricated average direction.

## 4 How everything can itself have an SDF definition

### 4.1 Representation and interpretation are different jobs

The goal that values, operators, expressions and time are each SDF-defined can be made mathematically literal. It does not require every such object to have the same meaning.

For each declared type  $\tau$ , let  $X_\tau$  be a collection of finitely describable values. Choose an injective representation

$$E_\tau : X_\tau \longrightarrow \text{Dir}(V_\tau), \quad \text{Dec}_\tau(E_\tau(x)) = x. \quad (5)$$

The decoder is defined on valid representations. Its interpretation is part of the declared calculus. An operator's *description* has an SDF representation, while the operator's *meaning* specifies the allowed input/output relationships between other representations.

#### In everyday language

The word “turn” and the word “chair” can both be written with letters. They do not therefore do the same thing. In the same way, an operator and an object can both have directional definitions without becoming interchangeable.

### 4.2 A constructive direction-only encoding

**Proposition 4.1** (Every finite definition record can be represented by an eigenaxis). Fix a finite alphabet and an unambiguous finite encoding of definition records. There is an injective representation of those records by oriented eigenaxes in a two-dimensional abstract carrier.

*Proof.* Convert a record  $s$  to a finite binary word. Prefix the word by a leading 1 and interpret it as an integer  $n(s)$ . This preserves leading zeros and distinguishes different word lengths. Set

$$E(s) = \left[ \begin{pmatrix} 1 \\ n(s) \end{pmatrix} \right]_+ , \quad L_s = \begin{pmatrix} 1 & 0 \\ n(s) & 0 \end{pmatrix}. \quad (6)$$

If  $E(s) = E(t)$ , then  $(1, n(t)) = a(1, n(s))$  for  $a > 0$ . The first coordinate gives  $a = 1$ , so  $n(s) = n(t)$  and  $s = t$ . Also  $L_s(1, n(s))^T = (1, n(s))^T$  and  $L_s^2 = L_s$ .  $\square$

The integer is carried by a coordinate *ratio*, not by the length of the representative vector. Scaling both coordinates does not change the encoded record. The two-dimensional carrier is a representational choice; it does not assert that a described object, mechanism or world is physically two-dimensional.

This proves that a finite description of an operator, an input, a clock rule, a pack, an observation or an expression can have the same native directional form. It does not prove that arbitrary finite data require only constant physical information. Distinguishing larger integers requires sufficient symbolic detail or directional precision. The finite-state information argument in Section 10 makes that boundary exact.

### 4.3 A common vocabulary, not an unexplained oracle

An expression such as  $\text{Then}(p, q)$  has its own encoded direction. So do Pack, Quote, Observe and  $T$ . Their semantics still have to be declared. Encoding a rule is not a substitute for saying what the rule means.

The starting assumptions are the ordinary logical reading of the definitions, the declared grammar, identity resolution and the interpretation clauses. These form the mathematical foundation of the calculus. They can themselves be described and quoted within it; storing their descriptions does not remove the need for a foundation from which those descriptions are read.

A definition can refer to its own identity or to another definition without containing infinitely many physical copies. Such references are finite expressions. Their meaning is supplied by the time or fixed-point conventions in Section 7, not by circular spelling alone.

### 4.4 One bit fits, but not every direction is one bit

In a one-dimensional real carrier,

$$E_{\text{Bit}}(0) = [-1]_+, \quad E_{\text{Bit}}(1) = [+1]_+.$$

There are exactly two oriented directions, so this is a genuine binary distinction. An unoriented line would merge them and lose the bit. In higher-dimensional carriers, additional directional distinctions can represent richer objects. The calculus therefore preserves the one-bit case without pretending that every possible definition contains only one bit of information.

## 5 Packing without merging the definitions

### 5.1 A lossless collection of individuals

Let  $I$  be a set of distinct definition identities. A finite pack has a registry of representations and an explicit collection of relationships:

$$\mathcal{P} = (I, \{\delta_i\}_{i \in I}, \mathcal{W}), \quad \delta_i = E(D_i). \quad (7)$$

Here  $\mathcal{W}$  specifies references and typed connections. Its finite descriptions can themselves appear as individually named definitions in the same registry. The notation separates them for readability, not because relationships require a different native substance.

**Definition 5.1** (Lossless packing). Packing is lossless when every constituent has a projection satisfying

$$\pi_i(\mathcal{P}) = \delta_i. \quad (8)$$

Two distinct definitions may have equivalent behavior while remaining different identities.

**Definition identity is not object-state identity.** One reusable transition rule may govern many instances. Sharing its description does not mean that the instances must share their changing states. Conversely, two references to one stateful instance must not be silently duplicated into independent instances.

#### In everyday language

A book can contain many recipes. You can retrieve each recipe, and two kitchens can use the same recipe without sharing one saucepan. Putting the recipes in one book neither mixes the ingredients nor makes the two kitchens identical.

## 5.2 Each direction keeps its own freedom of scale

For a pack of directional contents, the native mathematical object is a product

$$\prod_{i \in I} \text{Dir}(V_i), \quad (9)$$

with labels and relationships retained. Each representative  $v_i$  may independently be replaced by  $a_i v_i$ ,  $a_i > 0$ , without changing the contents.

This is generally *not* the same as taking one direction in  $\bigoplus_i V_i$ . In a single concatenated direction, only one common scaling is discarded. The remaining ratios between components would introduce information that the separate primitives never had.

For example, two separately represented directions  $[v]_+$  and  $[w]_+$  are unchanged when  $v$  is doubled. But the combined vectors  $(v, w)$  and  $(2v, w)$  are generally not scalar multiples. Treating those combined vectors as a single native direction would manufacture a relative magnitude between the definitions.

**A pack can still have its own directional description.** Apply the finite-record encoding to the description of the whole pack. That new SDF names the collection; it is not a vector sum of its members. Decoding it recovers the individual definitions and their independent equivalences.

## 5.3 The meaning of the packing symbol

D1 reserves the bare symbol  $|$  for definition-preserving packing:

$$\mathcal{P} \mid \mathcal{Q} = \text{Pack}(\mathcal{P}, \mathcal{Q}). \quad (10)$$

Identity collisions must be resolved by explicit sharing or consistent renaming. With that convention, packing is associative up to renaming and has an empty-pack identity. It introduces no interactions that were not declared.

A Boolean OR, a sequential operation and an alternative choice are separately defined operations. They can all have SDF descriptions, but they are not secretly the same operation because one printed symbol was used in an earlier discussion. This removes the earlier ambiguity without narrowing the possible operator vocabulary.

## 6 Operators, expressions and relationships

### 6.1 An operator is individually defined

A relation-type definition  $p : \tau \rightsquigarrow \sigma$  has an explicit meaning

$$R_p = \llbracket p \rrbracket \subseteq X_\tau \times X_\sigma. \quad (11)$$

Its directional representation is  $E_{\text{Code}}(D_p)$ . Its application semantics relate encoded inputs and outputs according to

$$E_\tau(x) \xrightarrow{p} E_\sigma(y) \iff (x, y) \in R_p. \quad (12)$$

No distance or inside/outside test appears in this definition.

A deterministic operation has one permitted output for each permitted input. A partial operation may have none. A nondeterministic relationship or design constraint may allow several. D1 keeps those cases distinct instead of forcing every definition to return one arbitrary answer.

### 6.2 The basic composition rules

For  $p : \tau \rightsquigarrow \sigma$  and  $q : \sigma \rightsquigarrow v$ , sequential composition is

$$(x, z) \in R_{p \circ q} \iff \exists y : (x, y) \in R_p \wedge (y, z) \in R_q. \quad (13)$$

The intermediate value is a witness connecting the two definitions. Independent composition is

$$((x, u), (y, v)) \in R_{p \otimes q} \iff (x, y) \in R_p \wedge (u, v) \in R_q. \quad (14)$$

Alternative choice between same-typed relations is  $R_{p \sqcup q} = R_p \cup R_q$ . Requiring both same-typed conditions is  $R_{p \cap q} = R_p \cap R_q$ . Identity has relation  $R_{\text{id}} = \{(x, x)\}$ . Rearranging named arguments is an explicitly defined permutation, not a change in what those arguments mean.

Each constructor has a finite description and therefore its own SDF representation under Equation (6). This gives **closure of the descriptive language**: composing defined relationships produces another individually representable definition.

#### In everyday language

One recipe says how to make a wheel. Another says how a wheel connects to an axle. A third says what a turning axle does. Composition joins the matching parts. Packing merely keeps the recipes together; composition says how they work together.

### 6.3 What a derivation establishes

A derivation records which definitions and intermediate values justify an outcome. It answers “why does this follow under these rules?” A valid derivation does not automatically show that an empirical rule is a correct description of nature. That requires observations and a stated physical interpretation.

Likewise, a relationship may describe all acceptable solutions without giving a short way to find one. The existential quantifier in Equation (13) states what a connecting witness must satisfy. It is not an assumption that the witness appears for free.

## 6.4 Equivalence and substitution

Definitions  $p$  and  $q$  are behaviorally equivalent for a declared observation family  $\mathcal{O}$  when every applicable observation gives the same answer for both. Exact relational equality,  $R_p = R_q$ , is a stronger and simpler sufficient condition for replacing one pure relation by the other in the constructions above.

For timed, stateful or resource-sensitive definitions, observations must include those features whenever they matter. Two expressions producing the same final symbol need not be interchangeable when one changes a shared state, takes a different modeled time or consumes a different modeled resource. The relevant equivalence must say what is preserved.

## 7 Time, self-reference and changing definitions

### 7.1 Time is a defined relation, not a compulsory fourth coordinate

Let  $\mathcal{S}$  be the collection of permitted situations and  $\mathcal{E}$  the collection of event descriptions. A time definition has semantics

$$R_T \subseteq \mathcal{S} \times \mathcal{E} \times \mathcal{S}, \quad s \xrightarrow{e}_T s'. \quad (15)$$

A situation can include active states and the currently active versions of definitions. An event says which change occurred. A history is a compatible sequence, or more generally a partially ordered collection, of such events.

Causal order  $e_1 < e_2$  says that the second event depends on the first. It does not by itself assign a number of seconds. A physical clock requires a separately defined calibration  $\theta$  and appropriate durations. A cyclic phase label alone also does not distinguish all elapsed cycles.

The earlier phrase “A is T” is retained as a possible explicit alias  $A \equiv T$ : a name referring to the time definition. It does not force time to be an opacity, a geometric dimension or a storage-channel interpretation.[\[H, P1\]](#)

#### In everyday language

Time first means “this can happen after that.” Seconds are something a clock adds. The rule for what happens next belongs in the same collection as the things that change.

### 7.2 Feedback with a past value

A rule such as

$$b_{k+1} = \text{NOT}(b_k) \quad (16)$$

is well-defined once  $b_0$  is supplied. Starting from 0, it produces 0, 1, 0, 1, .... By contrast, the simultaneous Boolean equation  $b = \text{NOT}(b)$  has no solution. The difference is the declared causal delay, not the appearance of the same symbol on both sides.

A finite acyclic dependency graph admits an evaluation order consistent with its dependencies. With specified inputs and pure deterministic rules, all such orders give the same results. Feedback can be unfolded to a finite event horizon when an initial state and a strictly causal delay convention are provided. A static fixed-point interpretation is another possibility, but it must supply existence and selection conditions rather than assume every loop has one solution.

### 7.3 Self-description and reflection

Quotation gives a directional representation of an expression:

$$\text{Quote}(e) = E_{\text{Code}}(e).$$

A declared interpretation relation specifies how a quoted expression relates an input to an output. The description of that interpretation relation can itself be quoted. This is **self-description**; it is not an automatic assertion that every statement the system can express is true.

Changing a rule is a transition of the form

$$(\mathcal{P}_k, s_k) \xrightarrow{e}_T (\mathcal{P}_{k+1}, s_{k+1}). \quad (17)$$

The event states which definitions changed, which old references remain meaningful and which versions were used to justify the outcome. Identity and versioning prevent a later rewrite from silently changing the meaning of an earlier derivation.

### 7.4 Learning as a particular kind of change

A learning process is not implied merely by storing rules as definitions. It is introduced by defining observations, a comparison criterion and an update relation on candidate rules. In schematic form,

$$(\text{rule}, \text{observation}, \text{criterion}) \rightsquigarrow \text{revised rule}.$$

All four descriptions can be SDF-defined. Whether the revised rule improves prediction or behavior is then a question answered by the defined criterion and independent observations, not by its use of the common notation.

## 8 Geometry, quantities and the source motifs

### 8.1 Measurements are separate definitions

The absence of native magnitude does not require an application to pretend that lengths, rates or amounts do not exist. It requires them to enter explicitly. A measurement definition can have semantics

$$\text{Measure}_{q,u}(x) = m, \quad (18)$$

where  $q$  names the measured property,  $u$  names the unit or reference convention, and  $m$  is a described value. The measurement rule and the value each have their own SDF representation. The representative length of either encoding is unrelated to  $m$ .

This also separates a *formal eigenaxis witness* from a *physical eigenmode*. A material model may define an operator whose eigenvalues have a physical interpretation, such as a stiffness-related quantity. That interpretation belongs to the material model and its units. The fixed eigenvalue 1 used in the encoding proof does not predict the material's response.

### 8.2 Circle, sphere, cone, apex and pyramid

The shape words in the source filenames can be retained without making every native definition a spatial scalar. The following are explicit D1 constructions, not transcriptions of the absent geometry source.[\[H, P1\]](#)

With a declared positive-definite inner product  $g$  (a Euclidean length-and-angle convention), center  $c$  and positive radius  $r$ , define a sphere surface by the relation

$$\text{SphereSurface}(x; c, r, g) \iff g(x - c, x - c) = r^2. \quad (19)$$

A circle is the corresponding surface relation in a declared plane. The center, radius, plane, measurement convention and surface relation all have individual definitions. The equality specifies a surface. An interior would require a separate predicate, for example  $g(x - c, x - c) < r^2$ ; it is not built into the primitive.

For a cone surface, define an apex  $a$ , an axis direction, the chosen inner product, an opening angle  $0 < \alpha < \pi/2$  and a forward parameter  $h \geq 0$ . Using that inner product only within this shape definition, let  $u$  be the unit representative of the axis. Then

$$x = a + hu + w, \quad g(w, u) = 0, \quad \sqrt{g(w, w)} = h \tan \alpha. \quad (20)$$

A finite cone adds a range for  $h$ . A pyramid can instead use named base vertices and triangular faces joining them to an apex. The apex itself is a specified point in that application; it need not supply a unique surface-normal direction.

#### In everyday language

“Point this way” does not tell you the size of a circle or the opening of a cone. Those are additional recipes. This preserves the direction-only primitive while still letting the collection describe complete shapes.

### 8.3 Direction alone does not determine a whole shape

At points away from a shared center, all concentric circles have the same radial direction  $[x - c]_+$ . They can nevertheless have different radii and boundaries. Therefore radial directions alone cannot distinguish those circles. A scale or equivalent relational constraint must be defined separately.

Similarly, a normal direction does not generally specify which side is to count as interior, and an eigenaxis does not identify a unique substance or behavior. The useful response is not to hide those missing choices inside the word SDF, but to make each choice another individually inspectable definition.

### 8.4 Log-polar organization without sneaking magnitude back in

In a declared two-coordinate chart, an angular label such as

$$\varphi = \text{atan2}(v_2, v_1)$$

is unchanged by positive rescaling of  $v$ . Its meaning depends on the chart; it need not be a physical angle in an application. Higher-dimensional carriers may require more coordinates or several charts.

A logarithmic scale label has a different status. It can be defined for an *additional* positive scale  $s$  relative to  $s_0$ :

$$\rho = \log(s/s_0). \quad (21)$$

Using  $s = \|v\|$  as an intrinsic property of a native direction would be invalid, because  $\log\|av\| = \log\|v\| + \log a$ . D1 therefore allows angular organization of directions and separately defined logarithmic scale descriptions, but it does not infer a native radius from an arbitrary vector representative.

A lookup collection or a binary search tree can organize named definitions. Neither organization is itself the semantics of a molecule, a route or a universe. The symbol  $\varphi$  is used here only as a chart angle; the missing source's exact interpretation of lowercase phi is not assumed to be an angle or the golden ratio.

## 8.5 Topology and the Klein-bottle motif

Topology can enter as separately defined identifications rather than as a demand that everything live in ordinary physical space. For example, take a square of parameter labels and impose

$$(0, v) \sim (1, 1 - v), \quad (u, 0) \sim (u, 1). \quad (22)$$

This is the usual edge-identification description of a Klein bottle. It supplies explicit neighbor and identification rules. A traversal across the twisted identification reverses the local orientation convention. A local coordinate arrow therefore needs an explicit transformation rule when it crosses that identification.

The mathematical object is not the same thing as a physical surface manufactured without intersection in ordinary three-dimensional space. Its value here is descriptive: the calculus can express different connectivities and local coordinate conventions without baking one global physical geometry into every primitive. The quotient construction above supplies the intended abstract identification directly; no image of a self-intersecting bottle is needed to define it.[\[R3\]](#)

## 9 Throughput: from definitions to resolved outcomes

### 9.1 Relational throughput

The word throughput can refer to what a system passes from input to outcome, or to a measured rate of doing so. D1 first formalizes the passage itself. This is the part determined by the definitions without introducing a particular physical device.

**Definition 9.1** (Through-image). For a relationship  $p : X \rightsquigarrow Y$  and an input collection  $A \subseteq X$ , define

$$\Theta_p(A) = \{y \in Y : \exists x \in A, (x, y) \in R_p\}. \quad (23)$$

The associated witness-preserving version retains the tuples  $(x, p, y)$  and, for a composition, the intermediate values and definition versions used in the derivation.

The plain through-image answers which outcomes are permitted. The witness-preserving version additionally records how each follows. Duplicated derivations of the same outcome are not automatically distinct useful answers.

**Proposition 9.1** (Composition law for throughput). For compatible relationships  $p$  and  $q$ ,

$$\Theta_{q \circ p}(A) = \Theta_q(\Theta_p(A)). \quad (24)$$

Also,  $\Theta_{p \sqcup q}(A) = \Theta_p(A) \cup \Theta_q(A)$ .

*Proof.* An element  $z$  belongs to the left side exactly when there are  $x \in A$  and  $y$  with  $(x, y) \in R_p$  and  $(y, z) \in R_q$ . The same witnesses state membership in the right side. The choice law follows directly from union of the defining relations.  $\square$

For independent inputs, Equation (14) additionally gives

$$\Theta_{p \otimes q}(A \times B) = \Theta_p(A) \times \Theta_q(B).$$

That product formula assumes independence. Correlated inputs or shared changing states require the actual coupled relation instead.

### In everyday language

A definition says which doors can lead to which next doors. Throughput means following those allowed connections until you reach an answer. Keeping a record of the doors you passed through explains the answer. Simply drawing more doors does not mean more questions have been solved.

## 9.2 Causal throughput without seconds

Choose a finite family of questions  $Q$ . Each question fixes the relevant definition version and specifies what counts as a resolved answer and what witness, derivation or complete search certificate is required. For a declared unfolding into causal stages, let

$$N_Q(k) = \#\{q \in Q : q \text{ has been resolved by stage } k\}, \quad (25)$$

counting each question at most once. For  $k > 0$ , define the stage-based rate

$$\tau_Q(k) = N_Q(k)/k. \quad (26)$$

This is a formal account of progress, not a hidden physical benchmark. The choice of elementary event and stage must be fixed before two derivations are compared. Declaring an entire difficult operation to be “one step” changes the stage model; it does not establish a real increase in throughput.

A witness may resolve an existence question. A universal claim, an impossibility claim or a global optimum may require more: a derivation covering all relevant alternatives, a valid bound, or another complete argument. A collection of promising candidates is not yet that certificate.

## 9.3 Causal depth, independent obligations and sharing

For a nonempty, finite, explicitly unfolded derivation, let  $W$  be the number of distinct required events and  $S$  the length of its longest dependency chain. The ratio  $W/S$  describes a potential for independent work under that event model. The largest group of pairwise independent events is another structural measure. Neither quantity is a rate in seconds.

Definition sharing can reduce repeated description and sometimes repeated reasoning. Suppose the same pure lemma is needed in many branches. Proving it once and citing it can replace many copies of the same derivation. This is a genuine logical benefit when the assumptions match. Reusing the result of a state-changing event under different states is a different claim and needs a separate justification.

**A description-saving example.** A rule of length  $\ell$  used  $m$  times costs roughly  $m\ell$  symbols when copied literally. A shared description plus  $m$  references of length  $r$  costs roughly  $\ell + mr$ , before the other connection information. This is a conditional accounting identity, not a universal compression ratio. It captures the reusable-definition goal without reference to a memory device.

## 9.4 Connecting throughput to a physical process

A physical interpretation adds a realization of the permitted transitions and an observation map. In the exact deterministic case, a suitable abstraction  $A$  would satisfy

$$A(F(h)) = T(A(h)) \quad (27)$$

for the physical states  $h$  in scope. Here  $F$  is the physical evolution over a defined interval and  $T$  the modeled transition. An approximate model instead specifies a comparison, tolerance and observation domain. A nondeterministic version requires the corresponding relational refinement condition.

Only after durations and observations are defined does a physical rate such as  $N_Q/\Delta t$  have a meaning. The calculus gives a vocabulary for that quantity; it does not determine its numerical value from the existence of directional definitions.

## 10 What follows mathematically, and the stronger questions

### 10.1 Properties established by the definitions

The direction quotient removes native representative length. The eigenaxis construction shows that every nonzero direction has a compatible witness. The finite-record encoding shows that values, operators and descriptions can share one direction-only form. Projection gives recoverable packing. Relational composition gives explicit semantics and the throughput composition law. Declared causal delay makes feedback meaningful.

These are mathematical consequences of D1's stated constructions. They do not depend on the tests in the earlier kernel companions, and those older tests do not establish this revised semantic model. The examples below are direct finite derivations of D1 behavior.

### 10.2 A finite representation still has an information requirement

**Proposition 10.1** (Distinguishable-record bound). A lossless representation of  $M$  distinct records requires at least  $M$  distinguishable representation states. A representation restricted to  $b$  binary choices therefore requires  $2^b \geq M$ .

*Proof.* Lossless recovery requires an injective map from records to representation states. A set of  $b$  binary choices has only  $2^b$  states. If  $M > 2^b$ , at least two records would share a state and could not both be recovered uniquely.  $\square$

This applies whether the representation is called a number, a direction, a word or an angle. The unbounded mathematical precision in Equation (6) does not create an unbounded number of distinguishable physical states in a fixed finite record.

### 10.3 Finite definitions can describe endless families

A short recurrence can describe an unbounded sequence. A surface equation can describe infinitely many points. A recursive grammar can describe arbitrarily large structures. This is a genuine extension beyond storing only an explicit list of instances.

But there are only countably many finite strings over a finite alphabet. Therefore not every arbitrary collection of infinitely many independent choices can be supplied by a finite description. Nor does a finite rule give all its answers at once. The distinction is between **specifying a family** and **resolving particular questions about it**.

#### In everyday language

“Keep adding one” describes counting forever. The instruction fits on a small card. It does not mean that every count has already happened, or that the card contains an arbitrary secret attached to every number.

### 10.4 What an eigenaxis-based solution theorem would require

A stronger research direction would use a problem’s structure to construct an operator whose selected eigenaxis represents a useful answer. For that to be a solution method, the construction must specify how the operator is obtained from the problem, how an appropriate axis is selected, and how a valid answer is recovered.

There must also be an account of operator size, precision, ambiguous eigenvalues and any search hidden in building the operator. Defining an eigenaxis after already knowing the answer proves representability, not discovery. An operator with one separate entry for every candidate may preserve a large search in its description rather than remove it.

This leaves a concrete possibility open: **derive structured operators from particular families of constraints, then prove that their recoverable directions solve those families under stated resource assumptions**. That would be an additional theorem, beyond D1’s general encoding and composition results.

### 10.5 Optimization is preserved as a question, not presumed solved

The travelling-salesman motivation survives in D1 as a relationship among places, permitted moves, visit requirements and an objective. A reformulation can expose symmetry or reusable constraints. A general reduction in problem complexity requires its own argument about construction, search, precision and verification. The standard distinction between describing a problem and establishing an efficient solution remains relevant.[\[R4\]](#)

## 11 Three complete examples in readable language

### 11.1 Example 1: a logical rule without little cubes

Represent false and true by the two directions  $[-1]_+$  and  $[+1]_+$ . Define the rule NAND by the following input/output relationship:

| First input | Second input | Output | Meaning       |
|-------------|--------------|--------|---------------|
| 0           | 0            | 1      | Not both true |
| 0           | 1            | 1      | Not both true |
| 1           | 0            | 1      | Not both true |
| 1           | 1            | 0      | Both true     |

The table is the readable meaning of one individually encoded operator definition. Its inputs and outputs also have individual directional definitions. For inputs 1, 1, the relation justifies output 0. No query about being inside a geometric region is involved.

For completeness, let  $N$  denote this rule. Then  $\text{NOT}(a) = N(a, a)$ ,  $\text{AND}(a, b) = N(N(a, b), N(a, b))$ , and  $\text{OR}(a, b) = N(N(a, a), N(b, b))$ . The stated table verifies each equality. Their compositions are again individually representable expressions. The packing symbol  $|$  still means packing, not NAND.

#### In everyday language

Every card can have the same kind of directional label. One card means “yes,” another means “no,” and another means “answer no only when both inputs say yes.” The rule is complete because its allowed answers have been defined, not because its arrow has a special length.

### 11.2 Example 2: a small travelling-salesman problem

Take four named places  $A, B, C, D$ . Define symmetric move costs, in a chosen abstract unit:

$$c_{AB} = 2, c_{AC} = 7, c_{AD} = 3, c_{BC} = 2, c_{BD} = 6, c_{CD} = 2.$$

These are invented numbers for this example. They are separate cost definitions, not magnitudes of the arrows naming the moves. Define an acceptable tour to start at  $A$ , visit each other place once and return to  $A$ .

Fixing the start and identifying a tour with its reversal leaves three tours:

| Tour                                                        | Cost derivation | Total |
|-------------------------------------------------------------|-----------------|-------|
| $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ | $2 + 2 + 2 + 3$ | 9     |
| $A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$ | $2 + 6 + 2 + 7$ | 17    |
| $A \rightarrow C \rightarrow B \rightarrow D \rightarrow A$ | $7 + 2 + 6 + 3$ | 18    |

The first tour is optimal *for this instance*, because all tours up to the stated symmetries have been covered. The definition pack contains the places, moves, costs, visit rule, return rule and comparison rule. The witness records the chosen tour and the exhaustive three-case comparison.

The through-image of the tour-feasibility relation contains the three tours. The optimum-selection relation reduces that collection to the least-cost result. Three feasible candidates and one proved optimum are different kinds of output. The axes supply a common representation; the explicit constraints and costs supply the problem's meaning.

### 11.3 Example 3: a structure that refills after damage

Take five named sites linked in a chain  $a - b - c - d - e$ . Define  $a$  as a permanent seed and give each site a separately defined on/off state. Let  $X_k$  be the sites that are on. Define

$$X_{k+1} = X_k \cup \{a\} \cup \{v : \text{a named link joins } v \text{ to some } u \in X_k\}. \quad (28)$$

Starting from  $X_0 = \{a\}$  gives

$$\{a\}, \quad \{a, b\}, \quad \{a, b, c\}, \quad \{a, b, c, d\}, \quad \{a, b, c, d, e\}.$$

If damage turns some sites off but leaves the seed and the link definitions intact, the same rule refills the chain. Every site is eventually reached because it lies at finite link distance from the seed. Once all five are on, the state is unchanged by another update.

This is a precise, limited form of modeled repair. It is not a claim that a living organism has been reproduced. The target connectivity is prescribed; the rule does not discover a new body plan. A physical version would also need material, energy, sensing and a way to perform the permitted transitions.

#### In everyday language

Five lamps know who their neighbors are. The first lamp stays on. Each dark lamp turns on when a neighbor is on. Turn a few off, and the same rule lights them again. That is a small, understandable example of recovery through local relationships and time.

## 12 Present-day possibilities, explained simply

### 12.1 What “enabled” means in this report

An application belongs to the possibility space when its individuals, relationships, changes and observations can be explicitly described in the calculus. That is **descriptive enablement**. A useful method also needs enough laws and data to answer the chosen questions. A physical realization additionally needs a demonstrated connection between those laws and the thing being modeled.

The applications in this section are **proposed uses of D1**, not reports of completed D1 products. They share one recipe: define the individuals; define permitted relationships; define what changes; define the question; preserve the explanation of the answer. Quantities are added only when the question needs them.

### 12.2 Routes, logistics and connected infrastructure

**Everyday picture.** Instead of keeping only a list of destinations, keep the destinations together with the rules for getting between them, the jobs to do there and what counts as a good trip.

In D1, each place, move, restriction, resource and cost is an individual definition. A road closure changes one relationship; questions that depend on it can be reconsidered without changing the meaning of every place. The same pattern can describe delivery routes, dependency chains, communication paths or evacuation *models*. A route witness explains which requirements a proposed answer satisfies.

The gain to investigate is clearer composition and reuse of related constraints. The least-cost answer still requires a valid comparison or proof. Merely replacing a list of cities with directions does not supply that proof.

### 12.3 Design by requirements rather than by one fixed shape

**Everyday picture.** Describe what a part must fit, carry and connect to before committing to exactly how it looks.

A design pack could contain a family of shapes, connection definitions, admissible material descriptions and independent requirements. A candidate design is an assignment satisfying those requirements. A change to one requirement defines a new question rather than requiring the conceptual description to begin again.

This is a route toward *inspectable inverse design*: searching for a mechanism or shape from its requested behavior. The formalization can keep the candidate, its assumptions and its acceptance argument together. Load, heat, flexibility, manufacture and tolerances require their own quantitative models. An axis naming a material is not a substitute for those properties.

## 12.4 Robot movement and interaction

**Everyday picture.** A robot needs to know not just where an object is, but what movements and interactions are allowed.

An object, grasp, joint configuration, contact condition and next action can each be separately defined. A movement plan becomes a chain of permitted transitions, with observations explaining when its preconditions hold. Direction-only primitives are compatible with orientation descriptions; actual positions, sizes, forces and durations remain separate application definitions.

The interesting possibility is one vocabulary linking a task, its geometric interpretation and the conditions for action. A physical robot still needs sensing, calibration, actuation and a model that remains accurate enough during movement. These are interpretation requirements, not properties that follow from the naming scheme.

## 12.5 Checking rules, protocols and engineered behavior

**Everyday picture.** Ask whether a set of rules can ever let the system reach a situation it was not supposed to reach, and keep the chain of events that demonstrates the answer.

D1's relations and time definitions can state allowed messages, resource ownership, dependencies and forbidden combinations. A counterexample is an explicit trace; a positive verification result needs an argument covering the stated model. This connects conceptually to constraint reasoning and satisfiability-modulo-theories methods, exemplified by the Z3 research system. Z3 is an external precedent, not an implementation or validation of D1.[\[R5\]](#)

The same descriptive pattern can support authorized analysis of software or device specifications. A bounded check speaks about its bound and assumptions. The absent source filename's "0 day" wording does not establish that the project already discovers unknown vulnerabilities.

## 12.6 Digital counterparts and changing scenarios

**Everyday picture.** Keep a model of a changing thing, the observations about it and several possible next chapters, without confusing the possible chapters with what actually happened.

D1 can separate observed definitions from assumed rules and hypothetical changes. A scenario is a versioned pack plus a transition history. Alternative assumptions produce alternative histories, and their resulting observations can be compared under one explicitly defined question.

Proposed uses include equipment behavior, resource networks and maintenance planning. The particularly useful feature would be traceability: the model can say which assumption caused a prediction to change. Predictions about the real object depend on the model's empirical adequacy. A consistent invented world can still be an inaccurate account of the real one.

## 12.7 Inspectable learned rules and structured knowledge

**Everyday picture.** A system changes its rules after seeing examples, but keeps the rules, examples and reasons for the changes distinguishable.

D1 can describe observations, a training objective, a rule family and a permitted update as individual definitions. This supports the possibility of learned components whose behavior can be inspected in a common descriptive language. It does not imply that every useful learned model has a short explanation comprehensible to a person.

A concrete neighboring research direction is learning networks of logic gates: Petersen and colleagues introduced a differentiable training approach whose resulting model uses discrete logical operations. The relevance is that learned rules need not be described only as unstructured numerical arrays; it is not evidence that D1 already reproduces that method.[\[R6\]](#)

## 12.8 Families of shapes, mechanisms and mathematical objects

**Everyday picture.** Keep the recipe for making all members of a family, not just a photograph of one member.

A recurrence, parameterized relation or shape constraint can describe many instances, including infinitely many mathematical instances. Individual parameters, exceptions and interpretation rules stay explicit. One definition of a joint or a growth rule can appear in many larger constructions without losing the identity of each use.

The possibility is a more unified library of *generative descriptions*: designs, logical expressions, route families and temporal behaviors can be composed in the same style. An infinite family remains a specification, not a completed answer to every question about every member.

# 13 The nature and mechanism branch, formalized

## 13.1 A common structure for a mechanism

The visible nature branch asks for literal mechanisms rather than a collection of analogies. Its source body is absent, so the mechanisms below are **new, explicit extensions**, supported where appropriate by independently checked research. No particular substance's identity or biological occurrence is inferred from its filename.[\[H, P1\]](#)

A mechanism pack can be written schematically as

$$\mathcal{M} = \mathcal{I} \mid \mathcal{R} \mid \mathcal{T} \mid \mathcal{O} \mid \mathcal{Q},$$

where  $\mathcal{I}$  names the individuals,  $\mathcal{R}$  supplies interactions,  $\mathcal{T}$  supplies change,  $\mathcal{O}$  supplies observations, and  $\mathcal{Q}$  supplies any required quantities and units. Each constituent, including each rule within a constituent, has its own directional definition. The packing sign does not itself cause an interaction.

### In everyday language

Naming the ingredients is not the whole recipe. You must also say which ingredients can interact, what that interaction does, what conditions it needs, and what you can observe afterward. Each part of the recipe gets its own definition.

### 13.2 Reaction networks and conservation

For a quantitative reaction *model*, define an amount vector  $n$ , a matrix  $N$  whose columns record changes in amounts, and reaction-rate functions  $a(n)$ . One possible continuous interpretation is

$$\frac{dn}{dt} = Na(n). \quad (29)$$

These are additional model quantities, not lengths of the native SDF directions. When  $Na(n) \neq 0$ , its directional class describes the direction of modeled change. It does not supply the rate; that remains in the quantitative interpretation. At a stationary state, the directional change is absent.

A useful exact check follows directly. If a row vector  $\ell^T$  satisfies  $\ell^T N = 0$ , then

$$\frac{d}{dt}(\ell^T n) = \ell^T Na(n) = 0.$$

Thus the defined quantity  $\ell^T n$  is conserved within this model. The algebra proves the invariant; identifying it with a measured chemical quantity needs the chemical interpretation. This is a concrete way to combine individually defined reactions into a mechanism with a checkable global property. Chemical reaction networks and their possible DNA-based realizations are studied explicitly by Soloveichik and colleagues.[\[R9\]](#)

### 13.3 Local interaction, transport and pattern formation

For named sites with concentrations  $c_i$ , an illustrative graph-based model is

$$\frac{dc_i}{dt} = D \sum_j w_{ij}(c_j - c_i) + f_i(c). \quad (30)$$

Here the links  $w_{ij}$ , the coefficient  $D$ , the local reactions  $f_i$ , the initial state and the observation rule are all separately defined. Multiple chemical species require the corresponding vector-valued quantities and transport coefficients. A specific pattern claim requires analysis of a specific choice of these definitions.

Turing's 1952 work analyzed how reaction and diffusion can destabilize an initially homogeneous state and produce spatial patterns in mathematical models. That gives a real mechanism family to study, rather than a claim that a direction-only primitive itself generates biological form.[\[R7\]](#)

In D1, the broader possibility is to compare different local rule packs while preserving the question: which interactions can produce a requested large-scale pattern, and under which conditions? The answer belongs to the composed mechanism, not to one isolated axis.

### 13.4 Adaptive networks inspired by living transport

A proposed transport model can give each link an individually defined state, let local use affect that state, and let a separate rule remove, retain or strengthen links. The adaptation rule and its tradeoffs must be stated. A cost criterion, reliability criterion and transport criterion need not agree.

Tero and colleagues studied network formation by *Physarum polycephalum*, compared relevant network properties with the Tokyo rail system, and developed a biologically inspired adaptation model. This is evidence for a concrete family of adaptive-network mechanisms, not for a general optimal travelling-salesman solver.[\[R8\]](#)

The D1 extension is to make such a mechanism inspectable as a pack of individuals, local changes and global observations. It could then be compared with a different rule family under the same stated objective.

### 13.5 Growth, repair, anisotropy and molecular identity

Growth needs rules for adding individuals and links. Repair needs a target or acceptance condition, detection of deviation, and transitions capable of reducing it. Directionally dependent material behavior needs preferred axes *and* separately defined response coefficients. These are distinct mechanisms even when they share a directional notation.

A molecular description likewise needs more than a silhouette or an axis: the proposed model must specify which molecular features matter, which interactions are permitted, and which conditions govern their consequences. A directional encoding of a substance's name alone predicts none of those consequences. D1's useful contribution is the ability to keep the necessary definitions explicit and composable rather than silently treating resemblance as a law.

## 14 Where present-day frontier research already connects

The following are **external research precedents, not D1 achievements**. They show that several ingredients in the possibility space have concrete mathematical or experimental counterparts. The connection proposed in each paragraph is descriptive: D1 could organize the individuals, rules, time and observations involved.

### 14.1 Learned growth and modeled regeneration: 2020

*Growing Neural Cellular Automata* demonstrated learned local update rules that grow specified images and, under the studied conditions, maintain or regenerate patterns after damage. These are constructed model systems, not living organisms.[\[R10\]](#)

The D1 connection is a pack containing an individual state definition for each agent, a shared local rule, separate neighborhood relationships, an update definition and a target observation. The earlier five-lamp example is much simpler: it gives a transparent repair-like rule, not a reproduction of the learned method.

## 14.2 Mechanical learning: 2024 and a 2025 theoretical extension

Li and Mao's 2024 study developed in-situ gradient acquisition for mechanical neural networks, studied learning tasks in simulations, and experimentally demonstrated gradient extraction in a three-dimensional printed mechanical system. This is more specific than a claim that a material has already learned arbitrary tasks autonomously.[\[R11\]](#)

Their March 2025 preprint analyzes a topological mechanical-network direction for classification and local learning. The reference here is explicitly its first preprint version, not an assumed later publication or general-purpose demonstration.[\[R12\]](#)

For D1, such models could separate orientation, coupling, response, observations and changes to the couplings. A mechanically meaningful eigenmode would require the mechanical operator and its quantitative interpretation, not just the universal encoding witness from Section 4.

## 14.3 Molecular rule realization: a constructive research route

Soloveichik, Seelig and Winfree gave a construction for approximating the kinetics of chemical reaction networks using DNA strand-displacement systems, with stated assumptions and supporting auxiliary mechanisms. The generality is a theoretical construction of a realization method, not evidence that every arbitrary network has been built experimentally.[\[R9\]](#)

The D1 connection would be a mapping from abstract reaction definitions to a chosen molecular mechanism, with an explicit agreement criterion for the observations. Producing such a mapping for a particular case is additional scientific work. A common descriptive syntax does not automatically make all substances interchangeable.

## 14.4 Shape-changing materials: 2016

Gladman and colleagues demonstrated biomimetic four-dimensional printing in which designed alignment within hydrogel structures controls anisotropic swelling and consequent shape change. Here "four-dimensional" names designed change after fabrication; it is not the primitive dimensionality of D1.[\[R13\]](#)

This is a concrete example of why preferred directions and quantitative response rules should remain distinguishable. A D1 description could keep local orientation, material response, environment and desired final shape individually specified. The paper's physical result does not establish a general material that can become any object on demand.

## 14.5 Architectural swarms: January 2026

The *Swarm Garden* research reported robotic modules combining local sensing, actuation and communication in reconfigurable architectural arrangements. Its case studies included adaptive window shading and interactive creative behavior. The authors describe it as a proof of concept for architectural swarms.[\[R14\]](#)

This is a particularly direct present-day connection to individually defined parts producing a coordinated larger behavior. A proposed D1 model would name each module, its neighbor relations, local state, update rules and collective objective. Scaling that to a self-maintaining building remains a separate scenario, not something established by the reported demonstrator.

## 14.6 Biomolecular structure prediction: 2024

AlphaFold 3 extended joint structure prediction to biomolecular complexes containing proteins, nucleic acids, small molecules and other specified components. Its reported task is structural prediction; that is not equivalent to a complete account of reaction kinetics, tissue behavior or treatment outcomes.[\[R15\]](#)

A D1 application could keep predicted structures, confidence-related observations, proposed interactions and experimental evidence as distinct definitions. The opportunity is better linkage between descriptions and hypotheses. It would not make a structural prediction a proven functional mechanism by changing its representation.

**What the research connections establish.** Local rules can be studied as mechanisms for growth, adaptation and collective behavior; some shape-changing and swarm systems have physical demonstrations. D1 offers a proposed way to describe these ingredients together. Their published results are not measurements of D1 and do not establish its superiority over their existing descriptions.

## 15 Far-future possibilities: the extended space

Every scenario in this section is **speculative as an integrated D1 application**. The aim is not to forecast a date. It is to show why the formalization can express the scenario, what additional mechanisms would make it meaningful, and what smaller result would genuinely support it.

### 15.1 Executable worlds whose laws are also objects

**ELI5 vision.** Imagine a world-builder in which the objects, the rules of the world, and the rule for what happens next are all individually inspectable recipes. Changing gravity in the story means changing a named law, not secretly changing every object.

D1 can describe this because rules, rule descriptions, states and time share the same representational form. A world is a pack with a chosen interpretation and a transition history. Different versions can branch from one history and answer questions about different assumptions. Invented topologies, unusual causal orders and nonstandard interactions can be defined explicitly.

The additional task is making the world's laws sufficiently complete and mutually compatible to resolve its intended questions. A world that depicts physics needs a physical model; a fictional world can deliberately use different rules. Rewinding a recorded model or constructing an alternative history is not physical time travel.

**A first discriminating result:** define a small world with two alternative law versions and prove exactly which observations remain the same and which change.

## 15.2 Digital organisms and self-repairing machines

**ELI5 vision.** A constructed thing notices that part of its organization is missing and follows its own local rules to restore a working form.

A D1 description can combine individual identity, neighborhood rules, a target condition, observations and feedback. It can describe replacement of damaged components as changes in the definition pack while retaining the identities and dependencies that should persist. Self-description can provide a record of the intended organization.

What remains to be supplied is the mechanism that turns detected differences into successful repairs. A physical system also needs replacement material, energy, sensing and actions compatible with its environment. Recovery to one prescribed form is narrower than open-ended life, reproduction or adaptation to arbitrary damage.

**A first discriminating result:** prove recovery from a clearly specified family of disruptions, then demonstrate that the observation map of a small physical or modeled system follows the same repair relation.

## 15.3 Swarms that assemble into useful machines

**ELI5 vision.** Many small pieces could organize into a bridge, a tool or a moving structure, with each piece following local rules rather than reading a complete picture of the final machine.

D1 can describe the identities of the pieces, connection compatibility, local observations, joining and separation events, and a collective acceptance condition. The pack retains individual states even when the assembly is treated as one named object. Shared laws need not mean shared state.

Additional requirements include reliable connectors, force and stability models, energy supply, communication constraints and rules for resolving incompatible local actions. Reaching a visual outline is not the same as creating a mechanically functional bridge. Different tasks may need different component capabilities.

**A first discriminating result:** show that a small collective reaches and retains one load-bearing configuration under explicitly defined component failures and disturbances.

## 15.4 Programmable matter and adaptive manufactured structures

**ELI5 vision.** A material could have a recipe for changing its organization when the situation changes: stiff here, flexible there, folded now, open later.

The calculus can separate preferred orientations, interactions, state-dependent response laws, environmental observations and a shape or function objective. A whole object can reference many local rules without erasing which local definition applies where. This supports a formal language for designing responsive behavior rather than only specifying a static final geometry.

The far-reaching version requires a physical mechanism with enough controllable states, reliable transitions, suitable energy and stable repeatability. An encoding of a desired shape is not a means of moving atoms into that shape. Molecular self-assembly, mesoscopic modules and responsive composites would require different realization maps.

**A first discriminating result:** one material system repeatedly reaches two specified functional states under a defined stimulus, with a quantitative model that predicts the observed transitions.

## 15.5 Virtual tissues and developmental design spaces

**ELI5 vision.** Rather than draw a finished tissue, describe how its parts communicate, divide, move and react, then explore which organizations those rules can produce.

D1 can represent cells or other modeled units, interaction networks, changes of identity or state, transport, mechanics and observations within one descriptive scheme. A developmental hypothesis becomes a specific pack of mechanisms whose consequences can be compared with data. Different biological scales need not be forced into one fixed spatial dimension or one universal type of variable.

The scientific requirements are substantial: adequate mechanisms, parameter identification, multi-scale agreement and experimental tests. A visual resemblance to a tissue is not proof of correct physiology. Predictions relevant to health would require their own rigorous validation; the formal calculus does not supply clinical conclusions.

**A first discriminating result:** predict one preselected pattern change caused by a specified perturbation in a well-characterized model system, then compare the prediction with an independent observation.

## 15.6 One description interpreted through different physical mechanisms

**ELI5 vision.** The same abstract game could be played with mechanical pieces, molecular interactions or optical components, provided each version obeys the same relevant rules.

D1 makes this possible as a *description goal*: give each realization an abstraction map and prove that its observed transitions agree with the specified relation. The agreement can concern only the requested outputs, not every microscopic detail. Section 9's realization equation states the central requirement.

Different media impose different noise, timing, coupling and resource conditions. A quantum extension, for example, would need additional definitions for complex state, relative phase, admissible evolution and observation probabilities. A real direction-only code may describe those definitions symbolically, but it does not thereby become the physical quantum state or acquire quantum advantages.[\[R16\]](#)

**A first discriminating result:** two different realized mechanisms satisfy the same small input/output and temporal contract within separately stated tolerances.

## 15.7 Automated mechanism discovery and theory revision

**ELI5 vision.** A research system proposes a recipe, predicts what should happen, checks an observation and changes the recipe when the evidence disagrees. It keeps the history rather than pretending the new recipe was always the truth.

D1 can encode hypotheses, experimental questions, observations, inference rules and revision rules as separate individuals. Self-reference lets the system describe and modify its own proposed models while retaining versioned derivations. Reusable submechanisms can be tested in more than one larger hypothesis.

What must be added is an effective search over hypotheses, an evidence model, suitable experimental access and a criterion for meaningful improvement. Consistency inside a description does not guarantee truth outside it. More expressive descriptions can also make model comparison harder when many hypotheses explain the same observations.

**A first discriminating result:** recover a hidden rule from a bounded family using a declared set of observations, then predict a new observation that was not used to choose the rule.

## 15.8 Designed topologies, adaptive habitats and new kinds of experience

**ELI5 vision.** A modeled environment could connect places in surprising ways, while a physical room could alter its configuration in response to how it is used. Both involve changing relationships, but they are not the same physical claim.

The topological branch can explicitly define unusual neighbor identifications or orientation changes, such as the Klein-bottle construction in Section 8. The architectural branch can define physical modules, local observations and configuration changes. D1 provides a common descriptive style while keeping the interpretation of each branch distinct.

A simulated doorway that connects distant places is a defined relationship in a model. A physical doorway requires a physically realizable arrangement. Likewise, a responsive façade is a component mechanism, whereas a self-maintaining habitat would additionally require integrated structural, energy, environmental and maintenance models.

**A first discriminating result:** prove the internal consistency of one nonstandard modeled topology, or demonstrate one well-defined collective adaptation in an architectural model. Neither result alone establishes physical manipulation of spacetime or a fully autonomous habitat.

## 16 What you actually have after this consolidation

Your resulting direction is a **calculus of individually defined directional eigenaxes, joined by explicit relationships and causal change, with optional interpretations for geometry, logic, mechanisms and alternative worlds.**

The original ambition survives in a more precise form. Things and operations can be represented in the same native form. Definitions can be packed without turning them into a single undifferentiated object. Time can be individually defined. Expressions and versions can describe other expressions and versions. Quantities can be present where needed without becoming the intrinsic lengths of the native primitives.

D1 supplies concrete mathematical constructions for these statements. It proves a direction-only representation of finite records, recoverable packing, and the composition law for through-images. It also separates a useful completed answer from a collection of candidates and an empirical realization from its abstract description.

The broad possibility space is therefore not “every application has now been built.” It is **one explicit descriptive basis on which many different application theories can be formulated, related and tested.** Which applications become useful depends on the additional laws, data, search methods and physical correspondences stated for them.

### In everyday language

You now have the rules for a common recipe book. Ingredients, instructions, links between recipes, changes over time and questions about the result can all have their own definitions. A recipe can describe another recipe, and many recipes can be kept together without losing their names. The exciting direction is how many kinds of mechanism the book can describe. Whether a particular recipe works is answered by its mathematics and, when it concerns the real world, by the world itself.

### Compact statement of the direction.

Individually define. Preserve direction without intrinsic magnitude. Pack without merging. Connect through explicit relations. Define time and observation. Derive outcomes with their witnesses. Add physical meaning only through an explicit interpretation.

## A Compact formal charter

This is the consolidated mathematical contract. It is a D1 construction from the requirements, not a quotation from an unavailable earlier source. The more detailed definitions and their examples appear in Sections 3–11.

**A1. Native direction.** For a declared real carrier  $V_\tau$ , an individual directional result is a class  $\delta = [v]_+$  with  $v \neq 0$ , under positive rescaling. No norm, boundary, interior or fixed physical dimension belongs to the primitive.

**A2. Eigenaxis witness.** An eigenaxis witness is an explicitly specified operator satisfying  $Lv = \lambda v$  with  $\lambda > 0$ . Witness data are distinct from native direction. An arbitrary direction admits a witness, but does not determine a unique physical operator.

**A3. Identity and representation.** Each record has a unique identity, a type and a finite expression. A declared injective encoding and decoder preserve the record's meaning. One construction is  $E(s) = [(1, n(s))]_+$ , with witness  $L_s = \begin{pmatrix} 1 & 0 \\ n(s) & 0 \end{pmatrix}$ . Typed interpretations distinguish a value from a rule or a description.

**A4. Individual-preserving packing.** A finite pack has indexed constituents and explicitly defined connections. It admits projections  $\pi_i(\mathcal{P}) = \delta_i$ . Independent rescaling of any constituent leaves the pack's meaning unchanged. Bare  $|$  is tagged packing, not addition, Boolean OR or application.

**A5. Relational operators.** A rule of type  $\tau \rightarrow \sigma$  has an explicitly defined interpretation  $R \subseteq X_\tau \times X_\sigma$ . Its description has its own directional representation. Sequential composition uses an intermediate witness; independent combination uses a product with separately identified inputs and outputs.

**A6. Explicit change.** A time definition relates an earlier state, an event and a later state. Feedback uses declared delay or an explicitly justified fixed-point interpretation. Physical duration is an optional calibrated quantity, not an intrinsic length or compulsory fourth coordinate.

**A7. Self-description.** Definitions may quote and refer to definitions through finite records and identities. Interpretation, update and version rules are explicit. A reference cycle alone entails neither a unique result nor a true proposition.

**A8. Observation and throughput.** An observation specifies the question and its resolution criterion. The through-image of  $R$  is  $\Theta_R(A) = \{y : \exists x \in A, (x, y) \in R\}$ . Witnesses preserve the justification. Causal rates count distinct resolved questions under a fixed event and stage convention.

**A9. Quantities and realization.** Measurements, units, physical laws, probabilities and realizations are additional individual definitions. They must satisfy the interpretation's consistency conditions. A representation of a physical claim is not itself evidence for that claim.

### A.1 The four central equalities

$$\begin{aligned}
 [av]_+ &= [v]_+ & (a > 0), \\
 \pi_i(\text{Pack}(\delta_1, \dots, \delta_m)) &= \delta_i, \\
 R_{qep}(x, z) &\iff \exists y : R_p(x, y) \wedge R_q(y, z), \\
 \Theta_{qep}(A) &= \Theta_q(\Theta_p(A)).
 \end{aligned}$$

The first removes representative length. The second preserves individuality. The third gives composed meaning. The fourth formalizes throughput through connected definitions.

## B Provenance, versions and the source ledger

### B.1 What was available and how it was used

| Record           | Availability                                                                 | Role in this consolidation                                                                                                                                                     |
|------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0               | Current user clarification; available in full.                               | Primary authority for the direction-only primitive, individual packing, non-device throughput and requested possibility space.                                                 |
| P1               | <i>JSFD Consolidation C1</i> ; supplied PDF, 35 pages, inspected.            | Previous account of history, relation semantics, geometric Boolean encoding and feasibility companion. Its distance/containment construction is not adopted as D1's primitive. |
| P2               | <i>What You Have Now: Synthesis 1.0</i> ; supplied PDF, 31 pages, inspected. | Previous broad synthesis and possibility mapping. Its implementation-centered framing is replaced, while useful distinctions about evidence and applications are retained.     |
| H                | Visible project conversation history and branch labels.                      | Evidence for the sequence of user requirements and attachment names; not a substitute for the bodies of those attachments.                                                     |
| Original sources | Geometry, routing and nature PDFs named in H; original bodies unavailable.   | Used only as provenance labels. Their absent equations, substance identities and detailed mechanisms are not reconstructed as quotations.                                      |

The two previous consolidation PDFs explicitly recorded their own source-access limitations. Reading them now makes their *actual constructions* available for comparison; it does not restore their missing original source material. The previous companion archives concern implementation feasibility and are outside D1's subject. Their reported tests do not certify the revised direction-only semantics.

### B.2 The visible revision trajectory

The initial geometry-oriented requests were followed by revisions asking for operators, expressions and time to be defined in the same notation. The V5 and one-bit branches emphasized compact individual definitions, logarithmic/polar organization and reuse. V6 explicitly resisted the interpretation that the project was merely a renderer or calculator, and connected the request to the ladybug/routing source. The nature branch sought extensions through literal mechanisms. The present clarification removes scalar distance and containment from the native definition.[\[H, P1, P2, P0\]](#)

These labels are used as historical references only. D1 is **a new consolidation edition**; it is not asserted to be a recovered frozen V5, V6 or other absent release.

### B.3 Original source-document names preserved as identifiers

The long filenames below are identifiers visible in the supplied history or C1's source ledger. Truncation is retained where the visible name itself was truncated.

**Initial geometry source, truncated:**

men T room a circle a sphere a cone an apex legenda a side view of the pyramid...

**Enhanced geometry source:**

men-T-room-a-circle-a-sphere-a-cone-an-apex-legenda-a-side-view-of-the-pyramid-lower-case-phi-log-encoded-polar-LUT-binary-search-tree-klein-bottl.pdf

**Ladybug and routing source:**

ladybug-anti-violence-tile-in-zaandam-at-the-train-station-got-me-to-thinking-about-the-travelling-salesman-NP-hard-problem-where-humans-try-to-solv.pdf

**Nature/mechanism source:**

bromofly and derivatives in nature insects bug finder 0 day signed distance field notations.pdf

The final name is reproduced solely for identification. It does not establish what “bromofly” denotes, a natural occurrence claim, a molecular structure, or an actual security result.

**B.4 Attribution and status of the mathematics**

**Originating direction and requested attribution:** Tom Klootwijk, NL200678942, 10-07-1990.

**D1 mathematical constructions:** oriented-ray semantics, the finite-record eigenaxis encoding, typed interpretation, individual-preserving packing, the specified operator algebra, causal throughput measures and the worked examples. These are explicitly supplied in this edition to make the clarified requirements precise. They are not represented as a transcription of missing original equations or as proof of historical priority.

**External research:** cited as evidence for particular mechanisms or neighboring capabilities. None of the cited authors is represented as endorsing D1, and their results are not relabeled as project results.

**C Plain-language glossary**

**Abstract carrier.** A mathematical setting in which a representation is defined; not automatically physical space.

**Oriented direction, or ray.** All nonzero representatives pointing the same way, with positive stretching ignored.

**Eigenaxis.** A direction preserved by a specified linear operator, up to its declared eigenvalue convention.

**Eigenvalue.** The scalar factor in an operator’s action on an eigenvector; it is not itself the direction.

**Definition record.** A named, typed expression saying what an individual definition is.

**Encoding and decoding.** The explicit correspondence between a readable description and its directional representation.

**Pack.** A collection whose individual definitions and connections remain recoverable.

**Relation.** A precise statement of which input/output combinations are permitted.

**Composition.** Connecting compatible relationships, with any required intermediate state identified.

**Witness.** Information that justifies a claimed relationship or answer; not necessarily a physical measuring instrument.

**Causal stage.** A declared level in a sequence or dependency ordering of events, not automatically a second.

**Through-image.** The outputs permitted by a relation for a specified collection of inputs.

**Resolved question.** A question for which the stipulated evidence or complete argument has been obtained.

**Interpretation.** The rules assigning meaning to descriptions; a physical interpretation additionally connects them to observations of the world.

**Self-reference.** A definition referring to its own description or to a collection that contains it, under explicit interpretation rules.

## D Sources and research references

**Project provenance** and **external primary sources** are separated below. External sources were checked for this report on 18 September 2026. Dates identify the work cited; they do not claim that it is the latest result in its area. The linked publisher or author-hosted records provide the source texts. Equations introduced as D1 constructions are derived in the report rather than attributed to those references.

### Reference register

- [P0] Tom Klootwijk. Current clarification in this conversation, 18 September 2026: direction-only individual SDF definitions, packed together; formal throughput and possibility space rather than device implementation. User-supplied text.
- [P1] Tom Klootwijk: *JSFD Consolidation C1 – What you actually have: Your goal, the formalization, the kernels, and the possibilities*. Supplied project PDF, 35 pages. In particular, its source ledger and mathematical appendix inform the comparison with D1. Local filename: Tom\_Klootwijk\_JSFD\_Consolidation\_C1.pdf.
- [P2] *What You Have Now: Klootwijk Field-Encoded Computing – Synthesis 1.0*. Supplied project PDF, 31 pages. Local filename: Klootwijk\_What\_You\_Have\_Now\_Synthesis\_1\_0.pdf. The previous descriptive title is recorded here for identification, not adopted as D1 terminology.
- [H] Visible project conversation history: initial geometry requests, V5 and one-bit branches, V6, and the biochemical/nature branch, including visible attachment names. User-supplied historical context, with some messages or names truncated.
- [R1] Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe and Steven Lovegrove. *DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation*. 2019. Used for the conventional scalar signed-distance meaning. [Author paper record](#).
- [R2] Gilbert Strang. *Lecture 21: Eigenvalues and eigenvectors*. MIT OpenCourseWare, 18.06 Linear Algebra, Spring 2010. Used for the standard operator/eigenvector relation. [MIT lecture and transcript](#).
- [R3] Allen Hatcher. *Algebraic Topology*. 2002. Author-hosted text, especially the opposite-edge identification of the Klein bottle on printed page 102; nonorientability and embedding distinctions elsewhere in the text. [Author-hosted book](#).

- [R4] Erik Demaine. *Lecture 16: Complexity: P, NP, NP-completeness, Reductions*. MIT OpenCourseWare, 6.046J Design and Analysis of Algorithms, Spring 2015. [MIT lecture record](#).
- [R5] Leonardo de Moura and Nikolaj Bjørner. *Z3: An efficient SMT solver*. 2008. External precedent for explicit constraint reasoning. [Author institution publication record](#).
- [R6] Felix Petersen, Christian Borgelt, Hilde Kuehne and Oliver Deussen. *Deep Differentiable Logic Gate Networks*. NeurIPS, 2022. [Author paper record](#).
- [R7] A. M. Turing. *The Chemical Basis of Morphogenesis*. Philosophical Transactions of the Royal Society of London B, 237, 37–72, 1952. Used for the reaction–diffusion pattern mechanism, not as a complete theory of biology. [Original paper hosted at Caltech](#).
- [R8] Atsushi Tero and colleagues. *Rules for biologically inspired adaptive network design*. Science, 327, 439–442, 2010. DOI: 10.1126/science.1177894. [Author institution publication and abstract](#).
- [R9] David Soloveichik, Georg Seelig and Erik Winfree. *DNA as a universal substrate for chemical kinetics*. PNAS, 107, 5393–5398, 2010. DOI: 10.1073/pnas.0909380107. [Author-hosted original paper](#).
- [R10] Alexander Mordvintsev, Ettore Randazzo, Eyvind Niklasson and Michael Levin. *Growing Neural Cellular Automata*. Distill, 2020. DOI: 10.23915/distill.00023. [Original interactive research article](#).
- [R11] Shuaifeng Li and Xiaoming Mao. *Training all-mechanical neural networks for task learning through in situ backpropagation*. Nature Communications, 15, 10528, 2024. DOI: 10.1038/s41467-024-54849-z. [Primary research article](#).
- [R12] Shuaifeng Li and Xiaoming Mao. *Topological mechanical neural networks as classifiers through in situ backpropagation learning*. arXiv:2503.07796, version 1, 10 March 2025. Preprint version explicitly cited. [Versioned author manuscript](#).
- [R13] A. Sydney Gladman, Elisabetta A. Matsumoto, Ralph G. Nuzzo, L. Mahadevan and Jennifer A. Lewis. *Biomimetic 4D printing*. Nature Materials, 15, 413–418, 2016. DOI: 10.1038/nmat4544. [Primary research article](#).
- [R14] Merihan Alhafnawi and colleagues. *Architectural swarms for responsive façades and creative expression*. Science Robotics, 11(110), January 2026. DOI: 10.1126/scirobotics.ady7233. [Author laboratory record and author-version paper](#).
- [R15] Josh Abramson and colleagues. *Accurate structure prediction of biomolecular interactions with AlphaFold 3*. Nature, 630, 493–500, 2024. DOI: 10.1038/s41586-024-07487-w. [Primary research article](#); [author institution abstract](#).
- [R16] Barton Zwiebach. *Lecture 12: Quantum Dynamics*. MIT OpenCourseWare, 8.05 Quantum Physics II, Fall 2013. Used for the distinction between physical quantum dynamics and a symbolic description of them. [MIT lecture materials](#).